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Sustainable Aluminium Brake Discs and Pads for Electrified Vehicles

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ABSTRACT: In recent years, there has been a paradigm shift in the automotive industry to phase-out internal combustion engines. This process requires the optimisation of every component of the vehicle for alternative drive systems. For instance, brake systems must be optimised to meet the performance requirements of electrified vehicles; enhanced corrosion resistance minimises non-exhaust particle emissions and negligible brake noise issues. The braking system of an electric vehicle relies mainly on regenerative braking, which implies that the issues of corrosion and brake noise (any vibration) will become prominent in the use of friction couples in electric cars. In addition, new particle emission regulations signify another hurdle that Original Equipment Manufacturers (OEMs) must fulfil in the coming years. However, these issues will become a great deal to contend with OEMs, while conventional friction brakes are still needed for safety reasons but are less used in electric vehicles. This investigation was motivated by the need to find an alternative brake-disc material that could satisfy the aforementioned challenges. This study highlights the prospects of a lightweight SICAlight rotor as a potential alternative to traditional grey cast iron (GCI) brake discs for future car applications with electrified propulsion. Cooperation between brake disc and brake pad manufacturers was initiated to evaluate the potential of the aluminium rotor with compatible brake pads to meet the requirements for electrified vehicle applications. The friction couples were investigated by running typical brake dyno and vehicle tests on a current production rear axle brake. The focus of this test was to confirm the overall performance of the friction couples in terms of wear, corrosion, non-exhaust emissions, durability, Noise, Vibration, and harshness (NVH). The results indicated a suitable braking performance with fast creation of a transfer layer at a stable level throughout the testing with low brake pad wear and virtually no disc wear, no corrosion issues, reasonable durability, and good NVH behaviour. In addition, a significant advantage is seen with lower particle emissions compared to conventional gray cast iron discs with the corresponding brake pad material. The friction couples (SICAlight rotor and pads) were also tested at the same level as the coated GCI brake rotor or even better in terms of particle emissions. With the support of regenerative braking and a refined brake rotor design, the low maximum operating temperature of the aluminium rotor can be optimised for brake applications. Friction couples are promising for application in battery electric vehicles (BEVs) to save weight, reduce unsprung mass, and reduce energy consumption for vehicle motion, eventually converting this into a longer driving range. This study provides additional application tests to confirm the suitability of the SICAlight disc as an alternative solution for braking systems in electrified vehicles. In summary, the investigation showed that friction couples (SICA rotor and compatible brake pads) are suitable to replace traditional brake friction couples for electrified vehicle applications in terms of sustainability, durability, particle emission reduction, corrosion resistance, and NVH issues.

KEYWORDS: SICAlight aluminium rotor, friction couples, dyno and vehicle tests, corrosion resistance, tribological performance, friction coefficient, non-exhaust emission reduction, weight reduction, and NVH issues.

1. INTRODUCTION

The introduction of electrified vehicles (EVs) has changed the landscape of the global automobile market, and owing to recent increases in fuel prices, EV penetration is moving closer to becoming the new standard. Electric vehicles are quickly replacing conventional automobiles in the global EV market. Sales of the plugin and battery electric vehicles surpassed 10.5 million in 2022, a rise of 55% from the previous year [1, 2]. This is reflected in the information presented in Figure 1. The share of electric vehicles in the automobile market slowly increased from 2013 to 2019, when a sharp increase was recorded. The recent exponential growth in the electric vehicle market share is partly due to national policies and incentives in various countries that bolster sales, a return to the exceptionally high oil prices seen in recent years, and the global adoption of EVs as a means of conserving the environment and reducing particulate matter emissions[3, 4]. As a result of the increased demand for electric vehicles, many original equipment manufacturers (OEMs) have focused on the manufacturing of EVs, as shown in Table 1. In 2022, a record-breaking 10.3 million EVs were manufactured by the top 15 automakers, accounting for more than 80% of the electric car output [5]. Owing to global initiatives to reduce carbon emissions and to transition towards a net-zero carbon future, the focus of numerous automotive industries has shifted towards vehicle electrification. This growing trend in electric vehicle production is expected to continue to achieve the United Nations' sustainable development goals - good health and

well-being, industry, innovation, infrastructure, climate action, and sustainable cities and communities.

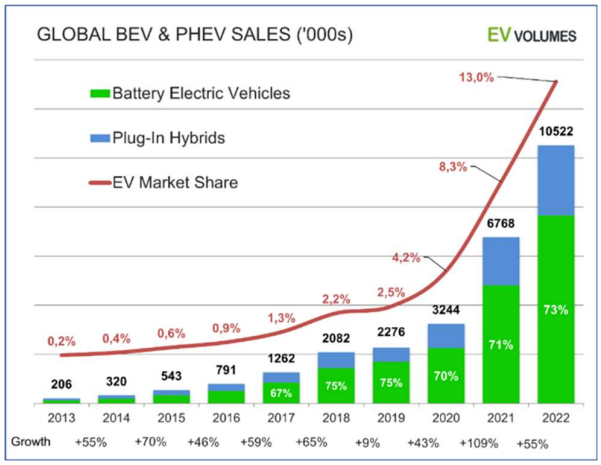


Figure 1 Global electric vehicle sales [2]

Table 1 Global electric vehicle production from 2021 to 2022 [5]

Rank	Company	2022	2021	Growth from 2021
1	BYD	1 858 364	598 019	211%
2	Tesla	1 314 319	936 247	40%
3	VW Group	839 207	763 851	10%
4	GM (incl. Wuling Motors)	584 602	516 631	13%
5	Stellantis	512 276	381 843	34%
6	Hyundai Motors (incl. Kia)	497 816	348 660	43%
7	BMW Group	433 164	329 182	32%
8	Geely Auto Group	351 356	99 980	251%
9	Mercedes-Benz Group	337 364	281 929	20%
10	Renault-Nissan-Mitsubishi Alliance	335 964	289 473	16%
11	GAC Group	287 977	125 384	130%
12	SAIC Motor Corp.	256 341	237 043	8%
13	Volvo Cars	253 266	220 576	15%
14	Chery Auto Co.	253 141	107 482	136%
15	Changan Auto Co.	245 555	105 072	134%
16	Other (41 companies)	1 927 211	1 326 262	45%

The transition from internal combustion engine (ICE) cars to electric vehicle transmission and driveline architectures involves different modes of operation, requirements, and challenges. When compared to fuel-powered vehicles, electric vehicles' original power systems, engines, and transmissions were replaced by a battery and high-power drive motor, which resulted in several changes to the vehicles' features and operations. One distinguishing trait of electric vehicles is the widespread use of regenerative braking, which allows for the conversion of kinetic energy into electrical energy while also applying negative torque to the driven wheels to slow down the vehicle [4, 6, 7]. The pattern of energy input and consumption for the braking system during vehicle deceleration changes because of this function, which could have a significant impact on some braking behaviours. Furthermore, the driving experience is quieter, and the acceleration performance is more potent, among other intriguing features. The key advantages of electric vehicles (EVs) include zero emissions during operation and reduced noise, vibration, and harshness (NVH) [8]. Regenerative braking accounts for more than 90% of EV braking and significantly changes the application area of friction brakes [9]. Friction brakes on EVs are utilised less frequently and with less force, which increases the likelihood that the disc surface will corrode over time and produce substantially higher emissions [10].

When corrosion of brake discs occurs, it can lead to other issues such as friction instability, generation of fine particles, and brake judder due to uneven thickness and brake disc deformation [10–13]. In addition, from a sustainability perspective and considering the categorical high vehicle masses of EVs, the question is raised as to whether it is necessary to drive around high unsprung and rotating masses with grey cast iron discs when brakes are rarely used.



Figure 2 A typical SICAlight brake disc

For electric vehicle brake discs to cope with the functional demands required, conventional gray cast iron (GCI) brake discs must be made more corrosion-resistant. This can be achieved by modifying the GCI brake disc via ferritic nitrocarburized (FNC) treatment, ceramic coating, or by changing the substrate material to obtain excellent corrosion and other required functional properties. Automakers and auto parts manufacturers have been strategising and searching for possible cost-efficient and sustainable solutions that would satisfy the above-mentioned challenges. However, aluminum-based SICAlight discs could be an alternative brake disc that would provide the required solution to the above-mentioned problems (Figure 2). The brake discs were comprised of silicon carbide particle reinforcement and an aluminium metal matrix to form an aluminium metal composite. Compared to traditional GCI discs, aluminium discs have advantages such as better wear resistance, increased thermal conductivity, excellent corrosion resistance, excellent heat transfer capability, no NVH issues, low particle emissions, and decreased weight [6, 7, 14]. Owing to the SICAlight disc's 50–60% lower weight than the GCI disc, the vehicle's ride and handling are enhanced, and the unsprung weight is decreased.

Regenerative braking is the primary distinction between braking in electric cars and vehicles powered by internal combustion engines. The motor reverses when an electric car brakes. Consequently, an electromagnetic force is generated that opposes the vehicle's forward velocity and restores the electrical energy in the battery while slowing it down. This function alters the pattern of energy input and consumption for the braking system during vehicle deceleration, which has a substantial impact on braking behaviour. Owing to regenerative braking, it is possible to drive an EV over long distances without a high load on the friction brakes. Rare use can accelerate the corrosion of tribological couples. It also permits the EV to apply brakes less frequently, which reduces the mechanical stress on the braking system [15, 16]. While this may

sound fantastic for brake life, it also poses new difficulties for electric vehicle owners. One of the vehicle parts that is most exposed is the brake component, such as the brake rotor surface, brake pads, and calliper sliding pins. These crucial components are exposed to weather, salt, and debris, which facilitate corrosion. For instance, if a car is left parked overnight in a humid environment, rust develops quickly on the friction surface of the rotor, as shown in Figure 3, because standard brake discs are prone to corrosion [6, 7, 11–13]. Naturally, moisture accumulates on brake pads during normal braking in any vehicle. This moisture is quickly evaporated in an ICE vehicle because it uses its brakes frequently and generates heat that dries off the moisture before it can corrode the pads. Unfortunately, this is not the case for an electric vehicle. According to reports [15, 16], EVs experience brake rust more frequently because of insufficient heat generation and lack of use, which both contribute to the corrosion of brake disc surfaces.



Figure 3 Corroded GCI brake disc mounted on a BEV

For the reasons previously mentioned, corrosion of brake friction surfaces heightens brake noise and other potentially uncomfortable elements. Corroded brakes may cause squeaking or grinding noises that continue even after repeatedly applying the brakes at greater speeds. However, this noise was not necessarily harmful. Extremely corroded brakes can become stuck, locked, or binding [11–13].

The expected limit value for brake particle emissions in the case of the EU7 regulations represents an additional challenge for the braking industry. Owing to their inadequate corrosion and wear resistance, conventional brakes are among the biggest pollutants on congested roads. A recent study conducted by Ghouri et al. [10] indicated that corrosion reduces the braking performance of GCI discs by decreasing the coefficient of friction, but it also markedly increases both the number and mass of particle emissions by at least a factor of 2 and up to 30 times when compared to the uncorroded brake disc under the same test conditions. The formation of numerous pits and cracks on the disc surface revealed that rust also affected the dimensional wear of the brake pads. The increase in the surface roughness of the corroded disc prevented the formation of a stable transfer layer and increased the amount of emitted PM.

Because of their high battery weight, EVs generally have higher vehicle weights than ICE vehicles. Therefore, automakers and suppliers are seeking new lightweight technologies to offset this weight. These technologies can directly contribute to improving vehicle efficiency in terms of kilowatt hours consumed per kilometre, as well as vehicle autonomy, and reduce environmental impact. Lightweighting can improve the strength, stability, and performance [17]. A lightweight brake disc reduces rotational inertia, improves dynamic behaviour, and reduces the unsprung mass, thereby improving suspension performance [18]. To address the various challenges highlighted, aluminium disc technology is considered a suitable alternative because of its low weight (up to 60% weight savings), stable friction level over a wide temperature range, low NVH, excellent wear resistance, and no corrosion. Thus, aluminium brake discs are most applicable to electrified vehicles.

This study evaluates the potential of lightweight SICAlight brake discs as a possible replacement for their conventional counterparts in future applications with electrified propulsion. In this investigation, the brake application performance, sustainability, durability, particle emission reduction, corrosion resistance, and NVH behaviour of SICAlight discs together with compatible brake pads were studied using a dynamometer and vehicle testing.

2. MATERIALS AND METHODS

2.1. Materials

The aluminium matrix composite brake disc used for this investigation was manufactured and supplied by Automotive Components Floby, Sweden. Blank aluminium discs were cast using the squeeze casting method and machined to the required dimensions. The composite was composed of AlSi9Mg0.6 (ASTM 359) and 20wt% SiC particles with a density of 3.2g/cm^3 and an average particle size of 18 µm. The brake disc under test was of a solid type with dimensions of 350 mm × 28 mm. To improve the surface properties of the machined aluminium discs, the friction surfaces of the rotor were chemically etched to ensure better bedding-in (quick transfer of the transfer layer) during the first contact with brake pads. Figure 4a shows the disc microstructure, and Figure 4b shows a sectional view of the protruded silicon carbide particles on the disc surface after chemical etching. Two temperature sensor holes were drilled into the discs via the embedded thermocouples before testing for data collection.

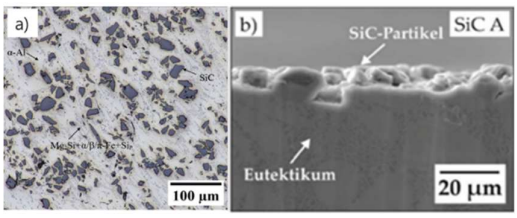


Figure 4 (a) Microstructure of the disc and (b) Protruded SiC particles on the disc surface after chemical etching [19]

Building on a production-released NAO-type friction material, a zero-copper formulation FER9701 was developed to fit the specific needs of the SICAlight brake disc. A suitable combination of lubricating, abrasive, and functional raw materials was selected to support the rapid creation of a stable friction layer and resistance to different duty cycles. In contrast to the materials frequently used in NAO formulations for gray cast iron rotors, a special combination of fibres was chosen to ensure stable friction levels and low wear rates. The sample brake pads were produced and provided by Federal-Mogul Bremsbelag GmbH, Germany.

2.2. Experimental Methods

To assess the brake performance, durability, NVH behaviours, PM emissions, and corrosion properties of the SICAlight discs and compatible brake pads, a series of dynamometer and vehicle test procedures were conducted, as described in the following sections. For an appropriate evaluation of the various properties and behaviours of the friction couple, the SICAlight disc with tailor-made brake pad cooperation has been agreed upon for testing according to typical requirements used to evaluate foundation brake systems for application projects.

In this study, an aluminium brake disc paired with an aligned pad material was tested in two different configurations. Table 2 shows an extract of the relevant vehicle parameters for the vehicle and dyno testing. The Volvo V90 configuration designed for advanced development was equipped with an internal combustion engine (ICE) propulsion at a test mass of 2300 kg. The car was equipped with 17” gray cast iron (GCI) discs on the front axle, and SICAlight discs on the rear axle. The rear axle dimensions of the disc types were Ø320x20 mm.

The Audi e-tron config with a default test mass of 2750 kg and an extra weight of 750 kg was used to simulate trailer operation and evaluate the brake performance during the alpine descent test. On the front axle, 18” GCI brake discs and ECE brake pads were installed. The rear axle was equipped with a solid-designed 18” prototype SiCAlight aluminium brake discs of 350x28 mm dimension and the corresponding brake pad Fer9701. The design of the aluminium brake disc is derived from the serial dimensions, considering the mechanical and thermophysical properties of the reinforced aluminium alloy.

Table 2: Overview of the test configurations

Config	Drive concept	Dimension [mm]	Dyno-Inertia [kgm²]	Vehicle weight [kg]
Audi e-tron	BEV	350x28	67	2750+750
Volvo V90	ICE	320x20	47	2300

2.2.1. Brake performance tests

To evaluate the brake performance, a standard friction test for inertia dynamometer testing was conducted using the Audi e-tron and Volvo V90 configuration in the standard AK-Master procedure similar to SAE J2522 [20] with reduced initial brake temperatures of 50°C instead of 100°C for most sections owing to the much faster cooling behaviour of Al discs and the lower average brake disc temperatures in EVs. The maximum initial brake temperature was limited to 300°C for the faded and high-temperature sections. As the creation of the transfer layer was identified as a key parameter for performance stability, interim stops were introduced during the early development phase to evaluate the transfer layer via photo documentation after bedding, fading, and completion of the entire test.

AMS100 Test

As a second performance test, a dyno simulation of the well-known auto motor & sport (AMS) magazine brake test (AMS100) with 10 applications for evaluation from 100 km/h to a standstill was performed on the dyno and the vehicle as a development tool. This is used as a reference for OEMs to adjust some details of such test procedures according to internal requirements.

Alpine descent

As a critical vehicle test for the evaluation of the brake disc and brake fluid temperature balance, the alpine descent test was performed at the Großglockner in Austria.

2.2.2. NVH durability and wear test

To assess NVH behaviour, a dyno test procedure was applied to an air-conditioned dyno with several noise matrices for variations in brake pressure, disc temperature, and velocity, as in the standard SAE J2521 [21] dyno brake noise procedure. Additionally, some adjustments were made to the low-temperature environment which better represents the predominant conditions for electric vehicles.

A Volvo V90 configuration was used for the NVH test during the developmental phase. GCI and SICAlight discs with the corresponding brake pads were mounted on the front and rear axles of the test car, respectively. An endurance test with combined driving conditions (55% countryside, 40% city traffic, 5% motorway) was conducted to confirm the overall durability, including NVH behaviour (Rear temperatures at max. 130°C, max. 180°C and max. 250°C) depending on the driving conditions. The test procedure was similar to the Mojacar test used by several car manufacturers for sign-off production activities. For data acquisition, the car was equipped with a JURID E3 system to collect noise-relevant data using microphones and accelerometers (frequency, SPL). After approximately 5,000 km of vehicle driving, the brake disc and pad wear were measured and used to estimate their life predictions.

2.2.3. Corrosion test

According to typical design verification plans related to passenger vehicle application projects, various corrosion test procedures are available from different OEMs and brake manufacturers, with a similar focus on characteristics such as stiction severity, disc cleaning capabilities, disc thickness variation (DTV), and friction performance. The dyno corrosion test procedure from Audi AG was used to simulate the corrosion behaviour of the friction couple. This test procedure was applied to a SICAlight disc with aligned pads and a reference GCI disc with compatible brake pads to evaluate the differences in their corrosion behaviour.

2.2.4. PM emissions test

Brake particle emission tests were conducted on a Link 3900NVH brake dynamometer equipped with an air-conditioned constant-volume sampling housing. The test brake was installed using a knuckle fixture. A Dekati ELPI+ aerosol analyser (cascade impactor) was used to evaluate particle emissions. The evaluation was performed according to the worldwide harmonised light vehicle test procedure (WLTP) [22] "Brake Cycle" developed by the International Working Group Particulate Matter Program of the UN ECE. The test runs were conducted in 2022 and do not correspond to the recently drafted method, but are helpful for back-to-back comparisons.

3. RESULTS AND DISCUSSION

3.1. Brake performance evaluation

Friction characteristics

One important finding of testing various friction material families was that not all materials were compatible with the SICA disc material; either the friction levels were not in the target range of μ≈0,40 after bedding or the creation of the transfer layer did not work, or the transfer layer was not stable enough to resist higher temperature duty cycles. Therefore, the developed copper-free FER9701 material showed fast creation of the target friction level, as shown in Figure , including a transfer layer established over only a few brake applications (see Figure 6). This result is at least as good as the copper-containing friction material on the SICAlight disc released for series application on the Scandinavian market approximately 10 years ago (Volvo V40). The absence of copper encourages the global application of friction couples.

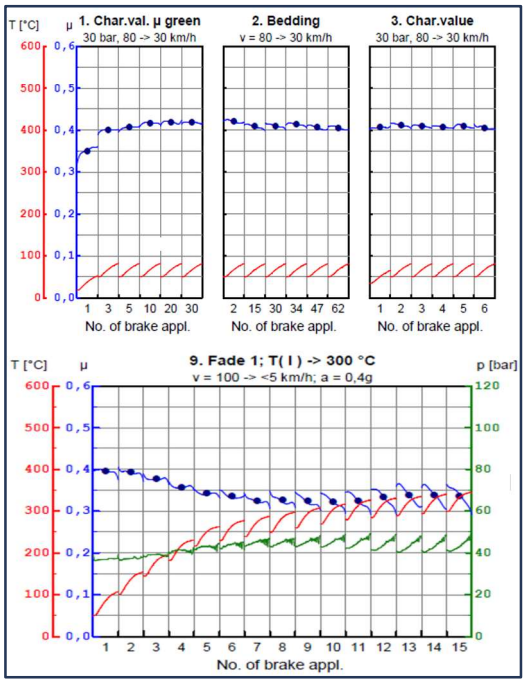


Figure 5: Extract of early and fade sections of the AK-Master dyno test of the tribo couple SICAlight and Fer9701 in the Audi or Volvo configurations

Tribological transfer layer

According to the photograph documentation after bedding, after the fade, and after the test (Figure 66), the visual aspect of the disc and pads showed very good coverage of the transfer layer on the friction ring.

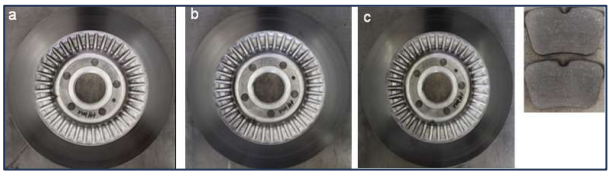


Figure 6: a) fast creation after bedding, b) resistance to duty cycle after fading 1, c) still good coverage after the entire dyno test.

Effect of surface treatment on friction characteristics

Another finding from the testing of friction performance is related to the surface treatment in the manufacturing process of the SICAlight brake disc. This topic was investigated using the Rear 18" brake. Without surface treatment, the friction levels are lower than the friction performance of a surface-treated disc, particularly at lower pressures (see Figure 7) which can occur more often with electric vehicles depending on the blending with the recuperation system. In addition, the formation of the transfer layer takes a longer time to occur with less stability and resistance to duty cycles on the untreated surface. Finally, there was a deterioration of the transfer layer after Fade 1, with no recovery until the end of the test, as shown in Figure 8. The surface roughness of new SICAlight discs was measured in a range of Ra = 2,05μm to 2,66μm. After the test, the Ra values of the surface-treated disc were found at Ra = 0,40 μm / 0,28 μm (inner/outer) whilst the surface roughness of the disc without surface treatment was measured at Ra = 1,30 μm / 2,07 μm (inner/outer-board).

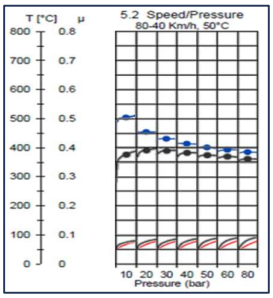


Figure 7: Example section from dyno friction test (AK-Master - the blue curves indicate the characteristics of the surface-treated disc, and the black curves represent the performance of the disc without surface treatment



Figure 8: Upper row: SICA disc with surface treatment, Lower row: SICA disc without surface treatment (AK-Master dyno test)

AMS test

The AMS100 is a well-known journal test for evaluating the stability of the brake performance. In Figure 89, the second five stops of the vehicle test are shown. Owing to the fully developed deceleration, the recuperation of the BEV was negligible. The condition of the rear axle SICAlight brake disc corresponded to the state after the Großglockner test program.

The results show an unobtrusive test with a constant CoF for the overall system at a reasonable pressure level. The front axle disc temperature reached $T_{max, FA} = 520$ °C, and $T_{max, RA} = 319$ °C on the rear axle at the end of the test. The friction surfaces of SICAlight exhibited an all-over formation of the expected transfer layer.

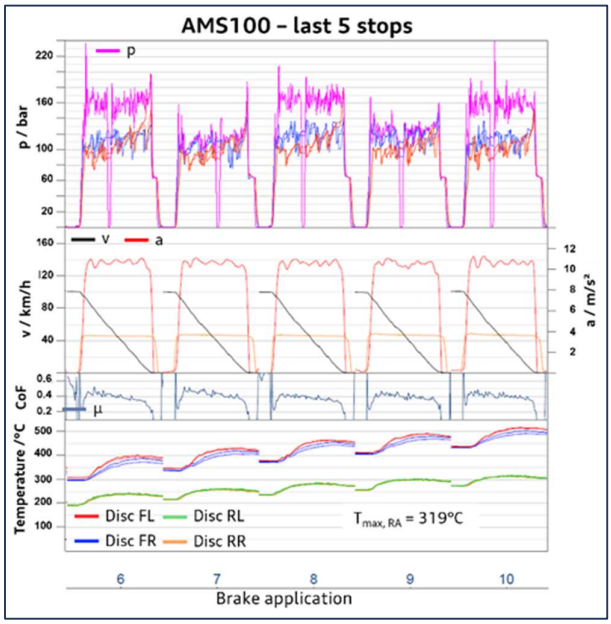


Figure 9: Evaluation of the second five stops of the AMS100 vehicle test with an Audi e-tron SUV equipped with GCI discs on the front axle and SiCAlight discs on the rear axle.

Alpine descent

The front axle was equipped with a standard GCI disc, whereas the rear axle was equipped with a SICAlight disc and tailor-made pad material (Fer9701). The temperature development of the brake discs and brake fluid during the Großglockner alpine descent is illustrated in Figure 810. The maximum brake disc temperature of the SICAlight disc was reached in the tourist descent at 321 °C, whereas the slow descent showed the maximum brake fluid temperature on the rear axle at 78 °C. Striking is the temperature development of the front and rear axles during a touristic descent.

The rear axle disc temperature remained at a certain level, while the front axle temperature increased further. The aluminium brake protects itself by reducing the CoF and shifting the brake power towards the front axle. This phenomenon became even stronger when the trailer operation was simulated by adding 750 kg to the trunk. Despite a significantly higher load, the maximum temperature did not exceed 350 °C. Figure 10 shows the rear axle brake disc after the Großglockner test program. The condition of the rear axle aluminium brake discs according to Figure 11 is good, with a smooth surface covered by a dark transfer layer in the same state after the AMS test and the Großglockner test program.

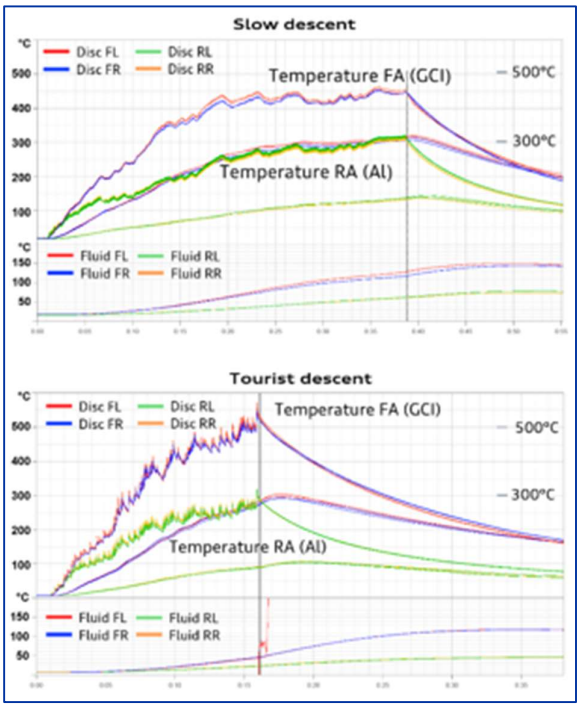


Figure 10: Temperature development of disc and brake fluid during slow and tourist vehicle descent at Großglockner.

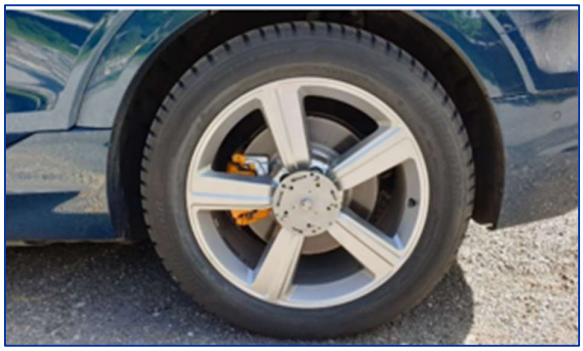


Figure 11: Image of the rear axle of the Audi e-tron equipped with the aluminium brake system after the Großglockner test program

3.2. NVH durability and wear evaluation

For evaluation of squeal noise behaviour the series Audi e-tron configuration of brake pad design features like chamfers or damping shim was carried over to the test pads and noise testing on dyno showed a low noise propensity overall at less than 1,9% occurrence with only high frequency (HF) noise at low sound pressure levels (SPL) around 80 dB of no distinct frequency, see Figure 12. The Volvo V90 endurance was tested over several thousand kilometres under various conditions, and good noise behaviour was observed, even during days of noise testing that provoked elevated disc temperatures of up to 130, 180, and 250°C on the rear, as described in Section 2.2.2. Figure 13 shows the driving and braking condition measurements at temperatures up to 250 °C on the rear axle SICAlight disc.

For noise events, an example of an SPL vs. frequency plot can be seen in Figure 14, with only high subjective ratings on a scale from 1 to 10. In general, the NVH experience with the SICA disc is positive with a low probability of generating noise. If high-frequency noise occurs, then it would be at a low SPL which rarely occurs.

After a relevant driving distance of approximately 5,000 km, the life prediction of the brake disc and pad was calculated based on the thickness losses from the brake disc and pad, as shown in Table 3, for the ICE propulsion vehicle where only friction brakes were used.

As the portion of usage of friction brakes is known to be reduced to less than 10% in electric vehicles under normal driving conditions, up to approximately 3 m/s ² deceleration stops [9], the life prediction is expected to be much higher than the numbers reported in Table 3. The deceleration capabilities of future EVs are expected to increase further for new vehicles that will come to the market over the next few years.

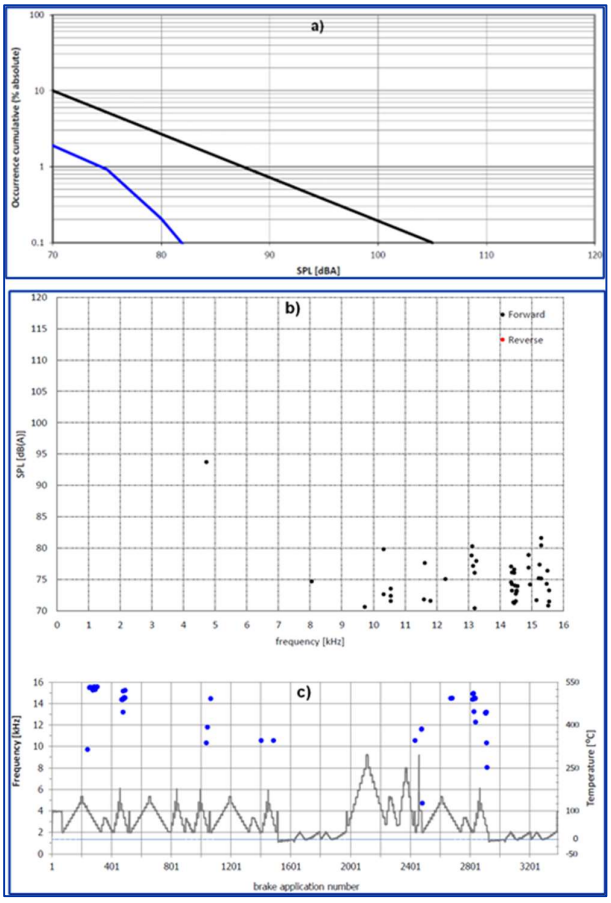


Figure 12: NVH dyno test results with Rear SICA-brake disc and pads in the Audi e-Tron configuration.
a) blue curve: Occurrence vs. SPL, black curve: Acceptance criteria guideline
b) Noise events indicated by dots: SPL vs. Frequency
c) Noise events indicated by blue dots: Freq. vs. brake application number.; black curve: Temperature profile

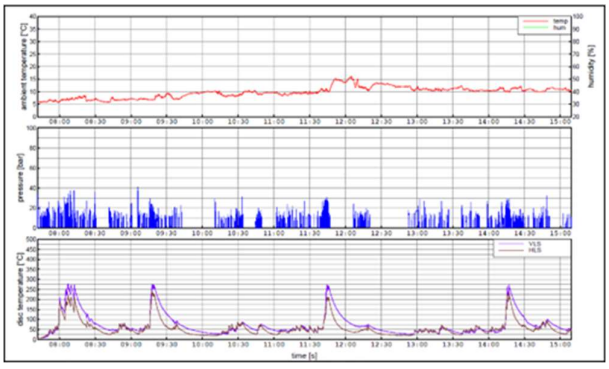


Figure 13: Measurements from one-day endurance test driving (top: ambient temperature; middle: brake pressure; bottom: front and rear disc temperatures) in Volvo V90 configuration.

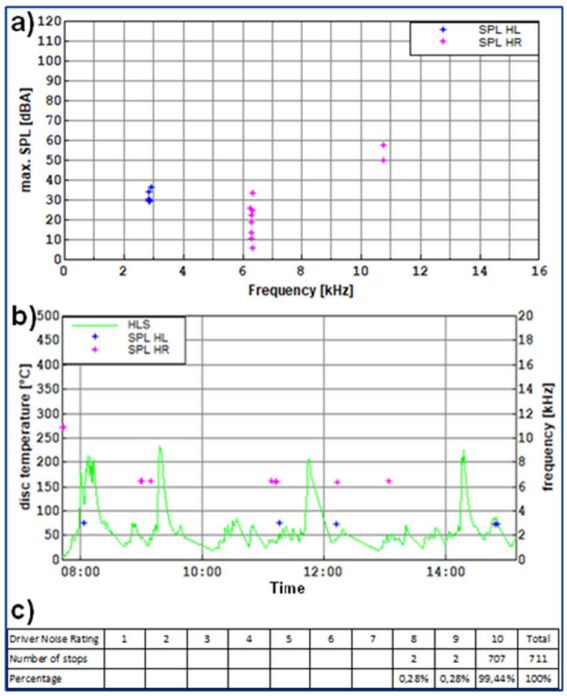


Figure 14: Rear noise data collected for one-day driving
a) SPL vs. Noise frequency of Rear left & right
b) Brake Temperature/Noise freq. vs. daytime
c) Subjective noise ratings for driver-noticed events.

Table 3: Disc and pad life prediction

Condition	Left	Right
SICA disc	No wear measurable	No wear measurable
Average brake pads life prediction	68 528 km	68 243 km

3.3. Corrosion evaluation

To understand the key difference in the corrosion behaviour of the SICA disc and the GCI discs in brake applications, a dyno test corrosion cleanability test was conducted. The main factors, such as the force released from stiction after the corrosion cycle and disc thickness variation (DTV), were examined as indicators of the cleaning capabilities of the friction couple. The significantly lower release forces reported in Table 4 confirm that almost no stiction issues occurred with SICA discs in a typical corrosive environment from a salt spray chamber. However, the release force for the GCI disc after the corrosion cycle was approximately 14 times that of the SICA disc, implying that higher brake stiction is expected with the traditional brake disc.

Table 4: Tangential release force of pads after the corrosion cycle

Condition	Inner	Outer
Current production pads on GCI-disc	972 N	2385 N
Tailor-made pads on SICA disc	70 N	77 N

Furthermore, Figure 15 shows that there is no creation of a DTV issue with the SICA disc pad system due to corrosion. The DTV measurement shown in Figure 15 indicates that the SICA disc DTV value decreased from a maximum of 13µm to 8µm after 10 brake applications, whereas that of the GCI disc decreased from the maximum value of 270µm to 8-10µm after 30 brake applications. However, the higher DTV observed in the reference disc leads to higher NVH issues with the GCI disc than with the SICAlight disc.

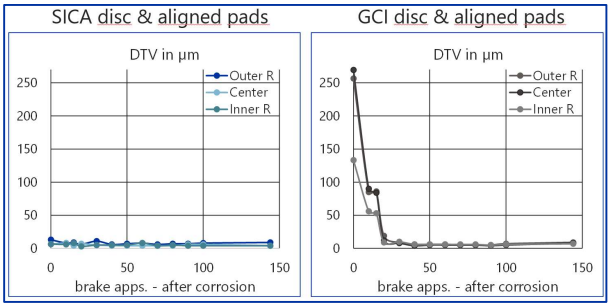


Figure 25: Comparison of DTV due to corrosion–cleaning capabilities

Finally, the visual appearance of the SICA disc and pad (Figure 16) after the corrosion test emphasised the durability and corrosion resistance of the SICA brake disc and pad friction couple. Because there is no corrosion issue with the SICA disc and pads, this friction couple would be the right approach to resolve brake corrosion issues with electric vehicles.

Here, it must be kept in mind that with corroded discs, depending on the environmental conditions, an electric vehicle is driven, and the particle emissions would increase when using GCI-discs with more aggressive pad material needed to clean again, which is not needed with SICAlight discs that do not have any issues with stiction and corrosion.

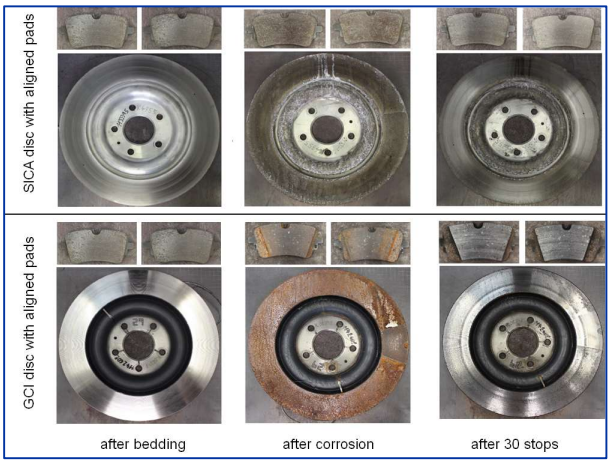


Figure 36: Comparison of test parts appearance through corrosion cleaning test

3.4. PM emissions evaluation

Figure 17 shows the brake particle emission results for the three different disc concepts paired with aligned pad materials. The gray cast iron disc (GCI) functions as a reference and shows PM10

emissions of 3,8 mg/km/brake. The SICAlight disc with the same disc dimensions of 350 × 28 mm combined with Fer9701 showed significantly lower emission values, with PM10 emissions of 0,85 mg/km/brake.

This represents a reduction of approximately 78%. This value is comparable to the measured emission values of a hard-coated gray cast iron disc. The results show that the SICAlight brake disc has the potential to reduce emissions to the same extent as a wear-resistant coating on gray cast iron. However, the use of a pad material with partial NAO properties and the formation of an extensive transfer layer on the aluminium brake disc play a crucial role in the PM emission behaviour of the tribological couple.

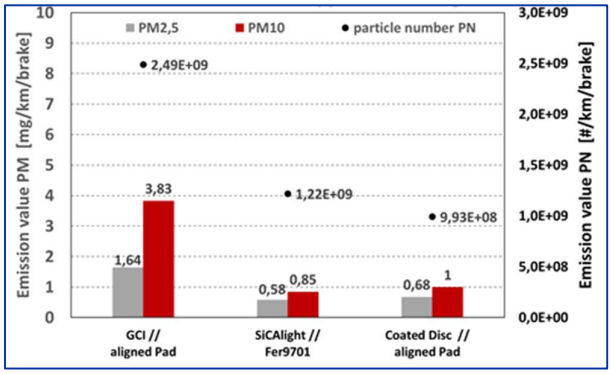


Figure 47: Comparison of brake particle emission values (dyno testing) for a rear axle application designed as a gray cast iron rotor, SiCAlight rotor, and coated gray cast iron rotor paired with aligned pad materials. Courtesy of Christopher Hamatschek, TU Ilmenau

4. CONCLUSION

This study investigated the brake performance of aluminium metal composite (SICAlight) discs using a brake dynamometer and vehicle tests with two different cars using specially developed compatible brake pads as tribological couplings. In addition, the corrosion cleanability of the SICA discs was evaluated using a dyno test, and the results were compared with those of the GCI reference discs. The results from the AK master and AMS dyno tests indicated a suitable braking performance with fast creation of the transfer layer at a stable level throughout the testing, with low brake pad wear and no disc wear, no corrosion issues, reasonable durability, and good NVH behaviour. A significant advantage is observed with lower particle emissions with SICA discs compared to conventional GCI discs with corresponding brake pads. The friction couples (SICAlight rotor and pads) were also tested at the same level as the coated GCI brake rotor or even better in terms of particle emissions. The study revealed that SICAlight discs with compatible brake pads generated 0,85 mg/km/brake PM10 emissions (about 78% reduction) compared to the reference GCI discs that produced PM10 emissions of 3,8 mg/km/brake. The corrosion test results indicated that there was a higher disc thickness variation in the GCI disc than in the SICAlight disc, which implied a higher propensity for NVH and PM emissions in the conventional discs than in the SICAlight disc. It can be inferred from this study that the corroded GCI discs would generate more particle emissions when combined with the more aggressive pad material needed to clean off the corrosion product created on the disc surface. It can be concluded that the SICAlight discs are free from stiction and corrosion issues, which makes them a potential brake component for electric vehicle applications. However, the application of SICAlight discs in EVs would benefit the brake system in terms of sustainability and environmental friendliness.

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